MILITARY SPECIFICATION SHEET

NUT (FASTENER), INTERNALLY THREADED, HEXAGON, REDUCED HEIGHT 125 KSI AND 200 KSI FTU, 450°F

This specification is approved by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the fastener described herein shall consist of this document and the latest issue of MIL-N-85353.

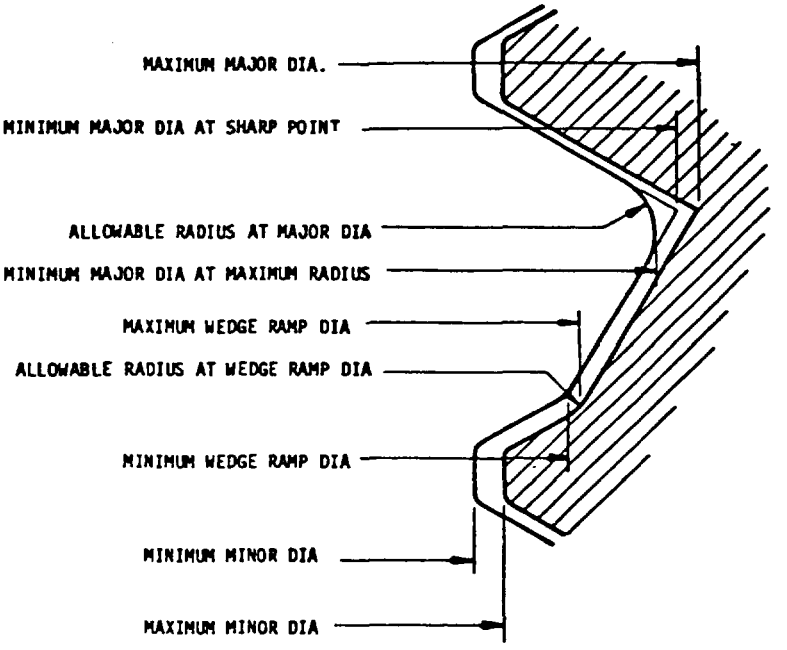


FIGURE 1. Internal Thread Form and Description.

FSC 5310

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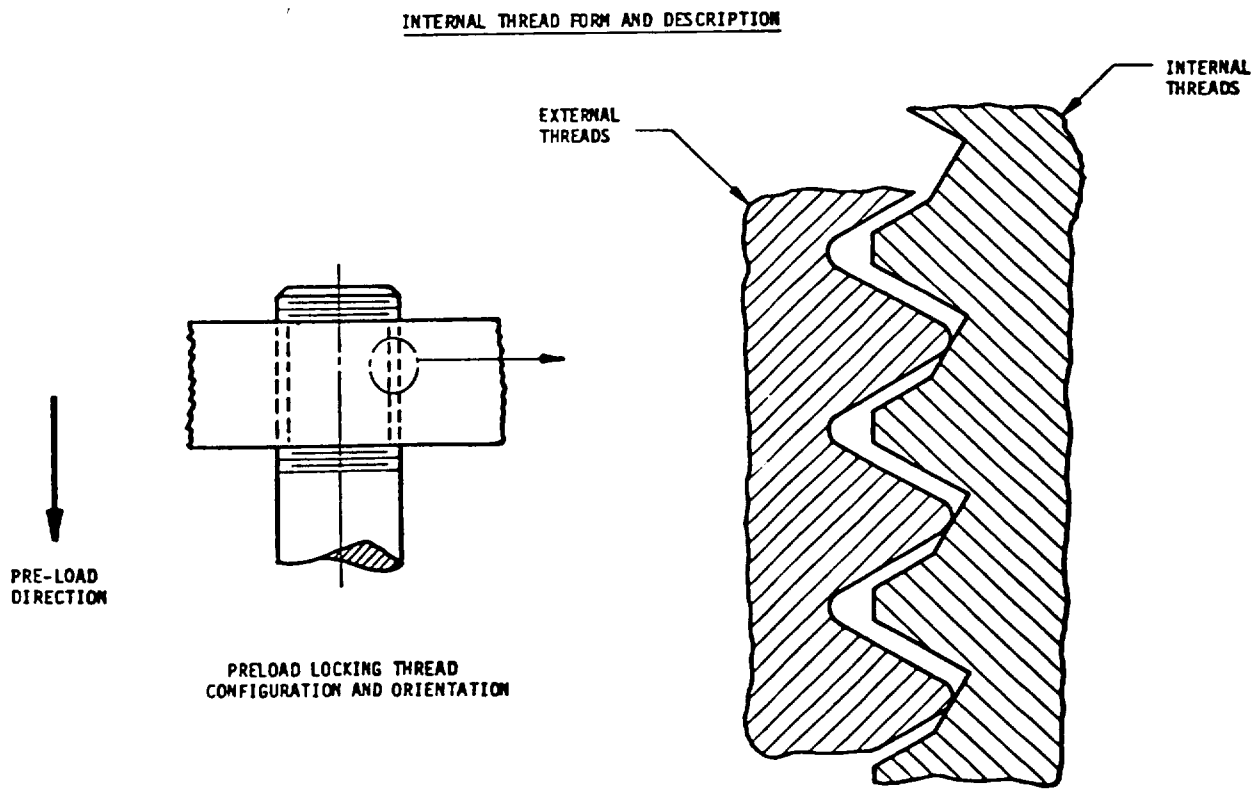


FIGURE 2. Locking Feature.

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THREAD DIMENSIONS

SIZE	AFTER PLATE DIMENSIONS									PRE-PLATE DIMENSIONS								
	MAJOR DIAMETER			MAX RADIUS AT MAJOR DIA (MMC)	WEDGE RAMP DIA		MAX RADIUS AT WEDGE RAMP DIA	MINOR DIA		MAJOR DIAMETER			MAX RADIUS AT MAJOR DIA (MMC)	WEDGE RAMP DIA.		MAX RADIUS AT WEDGE RAMP DIA	TAP DRILL DIA	
	MAX AT SHARP CORNER	MIN AT SHARP CORNER	MIN AT MAX RADIUS		MAX	MIN		MAX	MIN	MAX AT SHARP CORNER	MIN AT SHARP CORNER	MIN AT MAX RADIUS		MAX	MIN		MAX	MIN
.1120-40	.1161	.1135	.1129	.0010	.1029	.1004	.0149	.0942	.0877	.1169	.1143	.1137	.0013	.1033	.1008	.0119	.093	.089
.1380-32	.1427	.1399	.1391	.0013	.1263	.1236	.0209	.1157	.1076	.1435	.1407	.1399	.0016	.1267	.1240	.0179	.114	.110
.1640-32	.1673	.1645	.1643	.0003	.1517	.1489	.0213	.1417	.1336	.1681	.1653	.1651	.0005	.1521	.1493	.0183	.140	.136
.1900-32	.1930	.1902	.1901	.0001	.1751	.1722	.0235	.1675	.1596	.1938	.1910	.1909	.0004	.1755	.1726	.0205	.166	.162
.2500-28	.2536	.2504	.2502	.0003	.2340	.2308	.0276	.2229	.2152	.2544	.2512	.2510	.0006	.2344	.2312	.0246	.222	.218
.3125-24	.3169	.3133	.3129	.0005	.2943	.2907	.0302	.2799	.2719	.3177	.3141	.3137	.0008	.2947	.2911	.0272	.279	.275
.3750-24	.3795	.3758	.3754	.0005	.3552	.3515	.0343	.3418	.3344	.3803	.3766	.3762	.0008	.3556	.3519	.0313	.341	.337
.4375-20	.4429	.4388	.4382	.0008	.4165	.4124	.0370	.3970	.3888	.4437	.4396	.4390	.0011	.4169	.4128	.0340	.396	.392
.5000-20	.5055	.5013	.5007	.0008	.4779	.4737	.0392	.4591	.4513	.5063	.5021	.5015	.0011	.4783	.4741	.0362	.459	.455
.5625-18	.5682	.5638	.5632	.0008	.5387	.5343	.0437	.5166	.5084	.5690	.5646	.5640	.0011	.5391	.5347	.0407	.516	.512
.6250-18	.6308	.6263	.6257	.0008	.6001	.5956	.0459	.5788	.5709	.6316	.6271	.6265	.0011	.6005	.5960	.0429	.579	.575
.7500-16	.7567	.7517	.7510	.0010	.7241	.7192	.0497	.6977	.6892	.7575	.7525	.7518	.0013	.7245	.7196	.0467	.697	.693
.8750-14	.8826	.8773	.8763	.0014	.8469	.8416	.0541	.8152	.8055	.8834	.8781	.8771	.0017	.8473	.8420	.0511	.813	.809
1.0000-12	1.0088	1.0031	1.0018	.0018	.9699	.9642	.0616	.9298	.9189	1.0096	1.0039	1.0026	.0021	.9703	.9646	.0586	.927	.921
	1.1336	1.1277	1.1266	.0016	1.0923	1.0864	.0713	1.0539	1.0439	1.1344	1.1285	1.1274	.0019	1.0927	1.0868	.0683	1.054	1.048
1.2500-12	1.2587	1.2527	1.2516	.0016	1.2157	1.2097	.0773	1.1789	1.1689	1.2595	1.2535	1.2524	.0019	1.2161	1.2101	.0743	1.180	1.174
1.3750-12	1.3838	1.3777	1.3766	.0016	1.3410	1.3349	.0761	1.3039	1.2939	1.3846	1.3785	1.3774	.0019	1.3414	1.3353	.0731	1.305	1.299
1.5000-12	1.5090	1.5027	1.5016	.0016	1.4644	1.4581	.0821	1.4289	1.4189	1.5098	1.5035	1.5024	.0019	1.4648	1.4585	.0791	1.430	1.424

MMC --- MAXIMUM MATERIAL CONDITION

TABLE I. THREAD DIMENSIONS

SCREW THREAD DESIGNATION SHALL BE AS FOLLOWS: .2500 - 28 PER MIL-F-85353

.2500 - 28: THREAD SIZE; MIL-F-85353: SPECIFICATION NUMBER

ACCEPTANCE CRITERIA:

PRELOAD LOCKING THREADS ARE WITHIN DIMENSIONAL LIMITS WHEN THE GO GAGE (FIG. III) WILL ENGAGE ALL THREADS OF THE NUT WITH NO MORE THAN A SLIGHT DRAG AND THE HIGH LIMIT GAGE WILL ENTER THE THREADS NO MORE THAN TWO FULL TURNS. GO GAGE MUST ENTER DUAL LOCKING THREAD UNTIL ENGAGEMENT OF SECONDARY LOCKING DEVICE.

GAGE DESIGN:

THE GAGES ARE DESIGNED TO PROVIDE SURFACE CONTACT WITH THE UPPER FLANK AND THE RAMP MAXIMUM MATERIAL LIMITS ARE USED FOR THE GO GAGE AND MINIMUM MATERIAL CONDITIONS ARE USED FOR THE HIGH LIMIT GAGE.

PRELOAD LOCKING THREAD GAGES

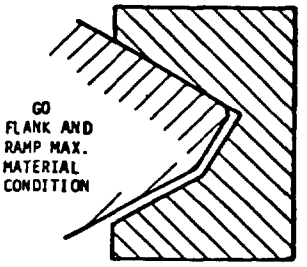


FIG. III

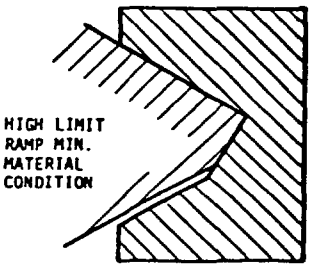


FIG. IV

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Patent notice. The screw thread trademark is Spiralock and is covered by Patent No. 4076064 expiring 2/28/95. The Government does not have a royalty free-license.

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Air Force - 11

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Army - AV
DLA - IS

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-N-85353/1 NUT (FASTENER), INTERNALLY THREADED, HEXAGON, REDUCED HEIGHT 125 KSI AND 200 KSI FTU, 450°F

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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